Charles Shaju

Embedded Systems Engineer

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SUMMARY

Embedded Systems Engineer with 2 years of experience in maritime robotics and IoT, currently pursuing an MCA to specialize in intelligent systems. Proven track record in developing Autonomous Surface Vessels (ASVs) and Underwater ROVs (UROV-SHM) using C/C++, ESP32, and Raspberry Pi. Expert in sensor fusion, real-time data systems, and UDP-based communication architectures for structural health monitoring.

WORK EXPERIENCE

Intern — 05/2026 – 05/2026
Jobin & Jismi — Thrissur, Kerala
- Educational Robotics & EdTech Development
- Robotics Web Platform & Rapid Prototyping
- Product Testing & Quality Assurance
- Smart Office Automation & IoT Integration

Embedded Systems Engineer — 08/2023 – 06/2025
i4 Marine Technologies Pvt Ltd — Pune, Maharastra
- Experience in Embedded Systems Development (C/ C++, ESP32, Raspberry pi)
- Worked on Underwater ROVs and Autonomous surface vessels (ASVs)
- Hands-on with Sensor Integration, Motor Control, LoRa, GPS, and real-time data systems

Robotics Intern — 01/2023 – 07/2023
Shristi Robotics — Thrissur
- Hands on building of Underwater ROV from Scratch
- Experience in 3D-Printing Technology
- Teaching various students on Mobile-Robotics

EDUCATION

MCA in Computer Application — 06/2025 – Present
Rajagiri College of Social Science — Kalamassery, Kochi

Bachelor of Vocational Studies(IT) — 08/2020 – 04/2023
Christ College Autonomous Irinjalakuda — Thrissur
GPA: 85%

HSS(Plus Two) in Commerce Computer Application — 07/2018 – 04/2020
St. George HSS Pariyaram — Thrissur
GPA: 97%

CERTIFICATIONS

MathWorks Computer Vision Engineer
- Issuer: MathWorks

Certificate in AI-Machine Learning Developer
Issuer: ASAP Kerala

PUBLICATIONS

Patent: Underwater Remotely Operated Vehicle (ROV) for Structural Health Monitoring — 12/07/2024
Indian Application No. 202241040473 A
UROV-SHM executes detailed underwater inspections of dams, tunnels, ship hulls, and bridge piers. Swappable batteries, fiber tethered telemetry, and AI-enhanced sonar/video analytics give real-time crack measurement, precise positioning, and mission resilience in turbulent conditions.

SKILLS

Embedded Systems
ADC/ DAC, Arduino, C / C++, Microcontroller Programming ESP32, PWM, STM32

Sensors & Actuators
Barometric Sensors, Depth, GPS, IMU, Motor Control, pH, Pressure, Temperature, Turbidity

Software Tools
Arduino IDE, Git, KiCad, Linux, Python, VS Code, Onshape, MATLAB

Communication Protocols
I2C, LoRa, RS485/ Modbus, SPI, UART, Wi-Fi

Systems & Platforms
Autonomous Surface Vessels, BlueROV, MAVLink, MAVProxy, Pixhawk, QGroundControl, Raspberry Pi, Underwater ROVS

Software Development
Python, Flask, Django, Java, SQL